

Game Genres

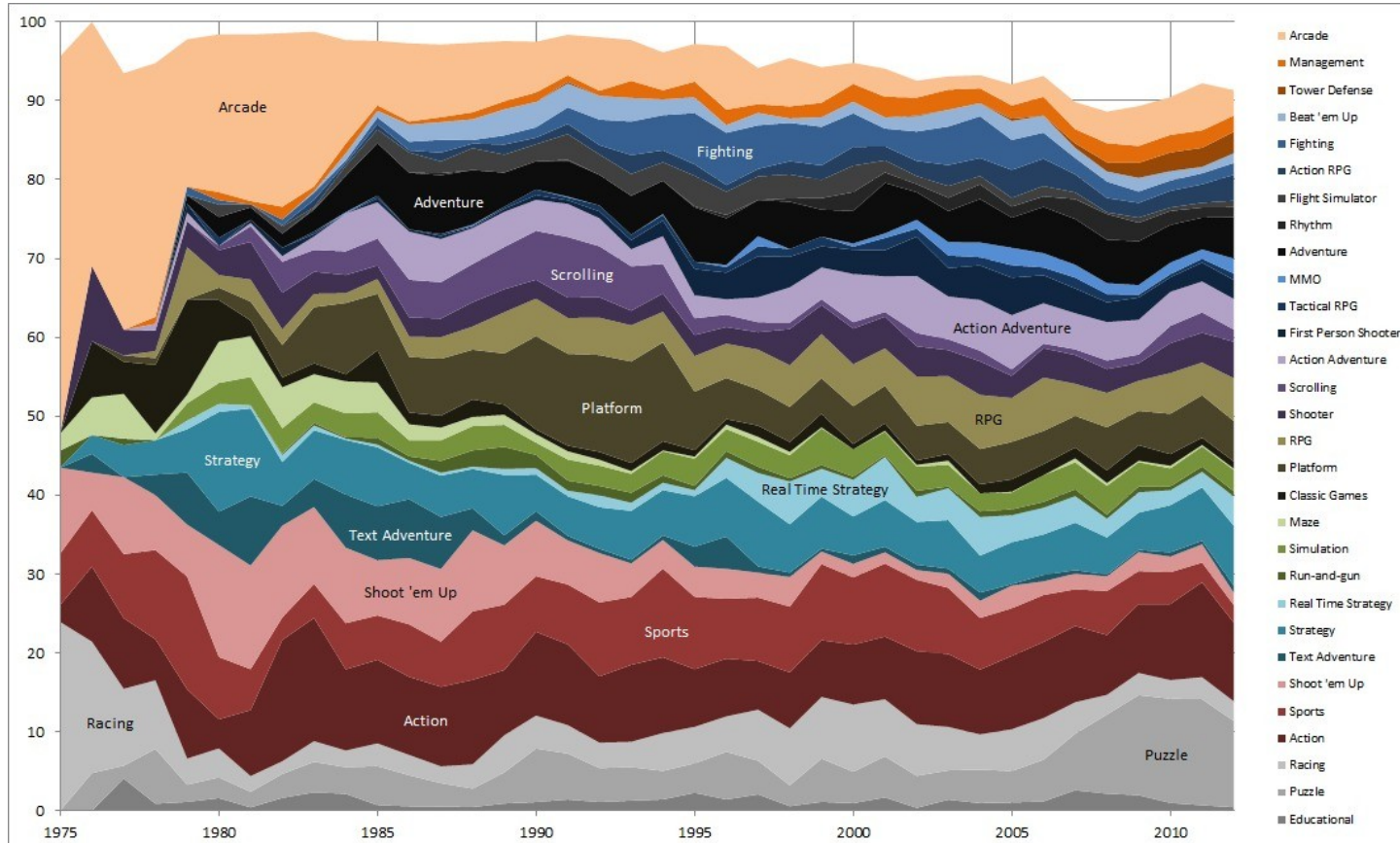


Image sources: <http://www.gameinformer.com/b/features/archive/2012/12/27/tracking-the-popularity-of-videogame-genre.aspx>

Conceptual Clarification



- Video game genres are used to categorize video games based on...
 - ... their gameplay interaction rather than visual or narrative differences,
 - ... their gameplay challenges (rules, goals, obstacles, etc.).
 - This categorization is (mostly) independent of the setting or game-world content, unlike other works of fiction such as films or books.



Point & Click Adventures



Action Role Playing Games

Image sources: http://store.steampowered.com/app/32360/The_Secret_of_Monkey_Island_Special_Edition/, http://store.steampowered.com/app/40330/Secret_Files_Tunguska/, <http://www.wallpapers4u.org/darksiders-war-horse/>, <https://thejesterscorner.wordpress.com/2012/11/10/zelda/the-legend-of-zelda-link-horse/>

Genres Discussion



Image sources: <http://gamerrevolution.co/the-gaming-genre-cheat-sheet/>

First Person Shooters (FPS) (1/3)



- Gameplay Characteristics
 - First person perspective
 - 3D environment (often hyper-realistic)
 - Combat is the primary design element
 - Combat is mostly gun and projectile weapon-based
 - Wide range of powerful hand-held weaponry
 - World traversal
 - Historically: Slow on-foot roaming in corridor-based worlds
 - Today: Usable vehicles like cars, boats and planes, large outdoor environments
 - Multiplayer gaming is often an essential part
 - Death-match, Capture the flag, etc.
 - Very common in eSports competitions
 - Implicit depiction of violence (often not youth-free)
 - Occasionally simple puzzles (e.g. lever mechanics or key cards)



Image sources: <http://weknowmemes.com/2015/04/first-person-shooter-logic-2/>

First Person Shooters (FPS) (2/3)



- Technological Characteristics
 - Extremely high (hardware) demands and very challenging to build
 - Many technological innovations arose from FPS games
 - Efficient rendering of large 3D virtual worlds
 - Highly optimized and carefully tuned rendering engine!
 - Space partitioning
 - Portal-based rendering
 - Various culling mechanisms
 - Very responsive camera and aiming controls
 - High fidelity animations of arms and guns (reload, jamming, ...)
 - High fidelity animations of non player characters
 - High fidelity artificial intelligence of non player characters
 - Forgiving player character motion and collision detection
 - Small-scale online multiplayer capabilities (typically up to 64 players)



First Person Shooters (FPS) (3/3)



Wolfenstein 3D



Half Life



Doom Eternal



Quake



Unreal Tournament



Call of Duty Mobile

Image sources: https://en.wikipedia.org/wiki/Wolfenstein_3D, [https://en.wikipedia.org/wiki/Half-Life_\(video_game\)](https://en.wikipedia.org/wiki/Half-Life_(video_game)), <https://www.theguardian.com/games/2020/mar/19/doom-eternal-review>, <http://wiki.quakeworld.nu/Quake>, <https://www.pcauthority.com.au/news/unreal-tournament-4-getting-closer-to-reality-384611>, <https://www.callofduty.com/mobile>

Platformers / Third-Person Games (1/3)



- Gameplay Characteristics
 - In platformers, jumping is the core gameplay element
 - Presence of a “jump” button or automatic jumping
 - High demands on the players dexterity
 - Many movement options, which may require precise execution
 - Moving platforms, ladders, ropes, trellises and other locomotion modes
 - Puzzle-like environmental elements
 - Third-person “follow-camera”
 - Stays focused on the avatar
 - Rotation is typically controlled by the human player
 - Often coupled with other genres
 - Action Adventures
 - Action Role Playing Games
 - Third-Person Shooters
 - Horror / Survival



Image sources: <http://nintendo.wikia.com/wiki/File:Super-mario-64-screenshot-lakitu-cameraman-big.jpg>

Platformers / Third-Person Games (2/3)



- Technological Characteristics
 - Comparable to First Person Shooters
 - Also extremely high (hardware) demands and very challenging to build
 - A great deal of emphasis is placed on the main character
 - Rich set of abilities and locomotion modes
 - High-fidelity full-body animations, with complex blending
 - Extended character collision system (compared to FPS), e.g.
 - Standing on narrow platforms
 - Grabbing onto ropes, ladders
 - Avoiding enemies and projectiles
 - Complex camera mechanics
 - “Intelligent” camera that stays focused on the action
 - Collision system to avoid clipping through foreground and background geometry



Platformers / Third-Person Games (3/3)



Donkey Kong



Tomb Raider



Demon's Souls Remake



Sonic the Hedgehog



Super Mario Galaxy



Mass Effect Series

Image sources: [https://en.wikipedia.org/wiki/Donkey_Kong_\(video_game\)](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)), <https://segmentnext.com/2019/04/13/shadow-of-the-tomb-raider-developer/>, [https://en.wikipedia.org/wiki/Demon%27s_Souls_\(2020_video_game\)](https://en.wikipedia.org/wiki/Demon%27s_Souls_(2020_video_game)), <https://www.videogamer.com/games/sonic-the-hedgehog-classic/screenshots>, <https://gamefaqs.gamespot.com/wii/915692-super-mario-galaxy/images/212>, <https://gamerant.com/mass-effect-legendary-edition-remastered-trilogy-mako-combat-multiplayer-me3-ending/>

Racing Games (1/3)



- Gameplay Characteristics
 - Primary task is driving a vehicle on some kind of race track
 - Players have to master the vehicle's controls and find the optimal racing line
 - Corridor-based or looped tracks, sometimes with short-cuts
 - Very fast travel speed (compared to on-foot roaming)
 - First- or Third-person camera
 - Many sub-genres
 - Simulation
 - Arcade-Racer
 - Street-Racer
 - Different types of competition
 - Time-based (who is the fastest)
 - Shoot at one another / collecting loot / other modes
 - Players often have the opportunity to tune and customize their vehicles



Racing Games (2/3)



- Technological Characteristics
 - Very detailed vehicle graphics and animation
 - High polygon counts and high-resolution textures
 - Damage and contamination model
 - Complex physics simulation
 - Friction, centrifugal force, bounce, crashes, wind, slipstream, etc.
 - Rendering of long race tracks
 - Level of detail (usually multiple levels)
 - Partitioning into sectors for optimization purposes
 - Rendering / visibility determination / path finding / etc.
 - Weather effects like rain, snow or sandstorms
 - Affecting visual appearance and driving physics
 - Complex artificial intelligence system for non-player opponents
 - One of the toughest challenges in the fields of game AI!
 - Sophisticated camera collision detection



Racing Games (3/3)



Lotus



Mario Kart



Gran Turismo



F-ZERO



Need for Speed, Juiced



DiRT Rally

Image sources: [https://en.wikipedia.org/wiki/Lotus_\(video_game_series\)](https://en.wikipedia.org/wiki/Lotus_(video_game_series)), <http://www.mobygames.com/game/wii/mario-kart-wii/promo/promoImageId,148402/>, <https://www.gtplanet.net/interlagos-is-coming-to-gran-turismo-sport/>, <https://gamefabrique.com/games/f-zero/>, <http://www.abc.net.au/tv/goodgame/stories/s1946102.htm>, <https://www.techspot.com/products/pc-games/dirt-rally-2.197547/>

Fighting Games (1/3)



- Gameplay Characteristics

- Two humanoid characters pummeling each other
 - They usually have equal physical strength
 - Mostly focused on close range combat
- Many attack options
 - Strong / weak, arms / legs, grabs, special moves, etc.
- Complex control schemes
 - Players have to master techniques like blocking, counter-attacking and combos
 - Specific button combinations execute special attacks
- Fights last over several rounds
- Very restricted roaming limited to some sort of ring or arena
 - The sub-genre “Brawler” or “Beat'em up” allows more character navigation
- Relatively static camera, focused on the action
 - Might zoom in/out or showcase special moves



Fighting Games (2/3)



- Technological Characteristics
 - Very detailed characters
 - High-definition models
 - High polygon counts
 - High-resolution textures
 - Skin shaders with subsurface scattering, sweat effects
 - Damage and injury model
 - High-fidelity animations
 - Physics-based cloth and hair simulation
 - Accurate hit detection
 - Head, Feet, Arms, Torso, ...
 - User input system capable of handling complex button combinations
 - Crowds, but otherwise relatively static backgrounds
 - No need for culling or world subdivision
 - No need to stream resources at runtime
 - Often fixed frame rate
 - Gameplay is often tied to the frame rate (e.g. windup or counter frames)



Fighting Games (3/3)



Street Fighter II



Virtua Fighter



Fight Night Round 3



Mortal Kombat



Tekken



God of War

Image sources: <http://mentalfloss.com/article/64071/15-things-you-might-not-know-about-street-fighter>, [https://en.wikipedia.org/wiki/Virtua_Fighter_\(video_game\)](https://en.wikipedia.org/wiki/Virtua_Fighter_(video_game)), <http://www.gamersglobal.de/galerie/841>, <https://www.mksecrets.net/index.php?section=kolumn&lang=eng&contentID=4177>, <https://www.fightersgeneration.com/games/tekken.html>, <https://www.gamesradar.com/god-of-war-review/>

Role Playing Games (RPG) (1/3)



- Gameplay Characteristics
 - The player controls the actions of a protagonist who lives immersed in the fictional world (hence the term “role playing”)
 - Originated in pen-and-paper role-playing games
 - Much of the same terminology, settings and game mechanics
 - Electronic medium removes the necessity for a gamemaster
 - Highly developed story and setting, divided into a number of quests
 - Large emphasis on story-telling and narrative elements
 - World exploration (story is often mapped onto exploration)
 - Items and Inventory
 - Finding loot and collecting it, such as clothing, weapons, and armor
 - Trading and management aspects
 - Experience and levels
 - Character's are gaining new skills, weapons, and magic (positive-feedback cycle)
 - Complex and challenging combat
 - Real-time or turn-based



Image sources: <http://presidentcaret.org/2017/10/18/role-playing-game/>

Role Playing Games (RPG) (2/3)



- Technical Characteristics
 - Extremely large game worlds
 - Subdivision into multiple areas (possibly connected by portals)
 - Excessive use of level of detail (long viewing distance)
 - Complex character and combat logic
 - Large set of attributes and dependencies
 - Mathematical equations often determine the outcome of a battle
 - Balancing of countless probabilities
 - Necessity for a very good random number generator
 - Extensive dialogue system
 - Individual decisions and nested conversation options
 - Large number of text lines and speech samples
 - Sophisticated character AI for party members and enemies
 - Challenging to save game progress (infinite number of states)
 - Serialization of the whole game world



Role Playing Games (RPG) (3/3)



The Legend of Zelda



Dungeon Master



Diablo II: Resurrected



Phantasy Star



Final Fantasy



Cyberpunk 2077

Image sources: [https://en.wikipedia.org/wiki/The_Legend_of_Zelda_\(video_game\)](https://en.wikipedia.org/wiki/The_Legend_of_Zelda_(video_game)), <https://www.dosgamesarchive.com/download/dungeon-master-ii-the-legend-of-skullkeep/>, <https://www.slashgear.com/diablo-2-resurrected-gets-some-good-news-for-fans-of-the-original-22660508/>, http://www.wikiwand.com/en/Phantasy_Star_IV:_The_End_of_the_Millennium, https://en.wikipedia.org/wiki/Final_Fantasy_VII, <https://www.rockpapershotgun.com/cyberpunk-2077-devs-say-they-ignored-signals-about-the-need-for-additional-time-to-refine-the-game>

Real-Time Strategy (RTS) Games (1/3)



- Gameplay Characteristics
 - Real-Time: does not progress incrementally in turns
 - Typically war games
 - Commanding an army and overwhelming the enemy
 - The Player only has indirect control of units
 - “Clicking and dragging” to select units
 - Giving orders and commands
 - Tactical decisions
 - Players have the ability to build new units and structures
 - Resource gathering and investment
 - Resources are usually scarce and hence limit construction and research
 - Research and development
 - Fixed viewing angle (or very limited)
 - Multiplayer gaming is often an essential part
 - Very common in eSports competitions



Real-Time Strategy (RTS) Games (2/3)



- Technical Characteristics
 - Each unit is relatively low-res so that large numbers can be rendered
 - Variable quantity of units must be supported
 - High demands on artificial intelligence
 - Steering behaviors and path finding for multiple units (avoidance!)
 - Tactical decisions and strategic planning
 - Grid-based world
 - Proper alignment of units and buildings
 - Optimized rendering and AI
 - Input features borrowed from desktop applications
 - Mouse input (click, drag, buttons, etc.)
 - Demanding and complex user interfaces
 - Orthographic rendering
 - Modern RTS games use perspective projection and a true 3D world



Real-Time Strategy (RTS) Games (3/3)



Dune II



Starcraft



Age of Empires



Command & Conquer



Warcraft III: Reforged



Homeworld Remastered

Image sources: https://en.wikipedia.org/wiki/Dune_II, [https://en.wikipedia.org/wiki/StarCraft_\(video_game\)](https://en.wikipedia.org/wiki/StarCraft_(video_game)), [https://en.wikipedia.org/wiki/Age_of_Empires_\(video_game\)](https://en.wikipedia.org/wiki/Age_of_Empires_(video_game)), <https://www.myabandonware.com/game/command-conquer-2r7>, <https://playwarcraft3.com/en-us/>, <https://www.rockpapershotgun.com/2016/05/23/homeworld-remastered-patch/>

Massively Multiplayer Online Games (MMOG) (1/3)



- Gameplay Characteristics

- Defined as any game that supports huge numbers of simultaneous players, usually all playing in one very large, persistent virtual world.
- Sub-genres
 - MMORPG (Role Playing Game)
 - MMORTS (Real Time Strategy)
 - MMOFPS (First Person Shooter)
- Gameplay experience is usually very similar to the respective small-scale multiplayer counterparts.
- Strong social aspects and extensive inter-player in-game communication
 - Inter-user chat or voice-over-IP services
 - Quests often require players to work together
 - Various external communication platforms
 - Highly addictive (symptoms comparable to nicotine or alcohol addiction)
- Individual financing models
 - Regular subscription fees
 - Free-to-play elements

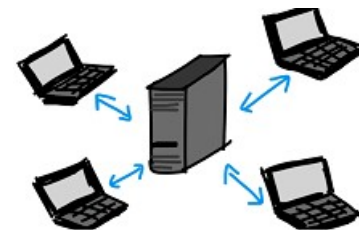


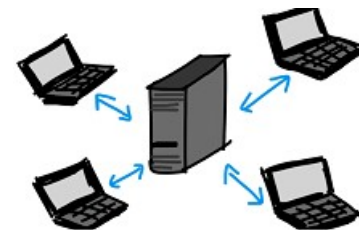
Image sources: <http://electronicsguide4u.com/peer-to-peer-network-and-client-server-technology-computer-server-based-network/>

Massively Multiplayer Online Games (MMOG) (2/3)



- Technical Characteristics

- At the heart of every MMOG is a battery of powerful servers.
 - Maintaining the authoritative state of the game world
 - User sign-in and sign-out management
 - Providing communication services
 - Payment system (billing and micro-transactions)
 - Statistics and data evaluation
 - User database
- Graphics fidelity is almost always lower than in its non-massively multiplayer counterparts.
 - Huge world sizes
 - Large numbers of supported users
 - Deliberately keeping hardware requirements low
- Character editor and other customization options
 - Models are made of multiple components (head, hair, torso, etc.)
 - Modifiable scripts and data files (e.g. Lua or XML)



Massively Multiplayer Online Games (MMOG) (3/3)



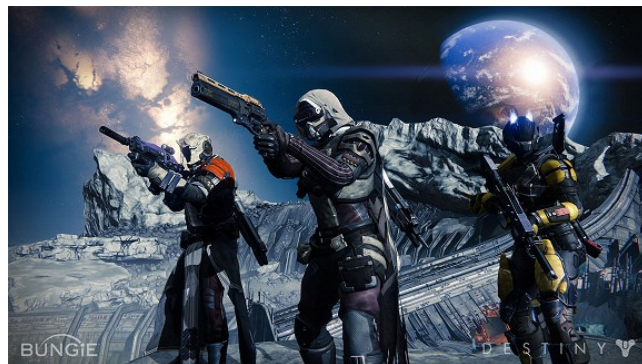
Everquest



Final Fantasy XIV



World of Warcraft



Destiny

Image sources: <https://en.wikipedia.org/wiki/EverQuest>, <https://heavy.com/games/2013/05/final-fantasy-xiv-a-realm-reborn-game-trailers-and-mmorpg-ps3-pc/>, <http://uk.businessinsider.com/world-of-warcraft-in-2015-2015-6?IR=T>, <https://kotaku.com/all-the-ways-activision-is-avoiding-saying-what-destiny-1654985376>